

CS & IT ENGINEERING



COMPUTER ORGANIZATION AND ARCHITECTURE

Instruction & Addressing Modes

Lecture No.- 01

By- Vishvadeep Gothi sir



Recap of Previous Lecture



Topic

Micro operations



Topics to be Covered



Topic

Instructions

Topic

Opcode

$$\frac{x * 2}{\downarrow}$$

one time left shift

$$\begin{aligned}(11)_2 &= 3 \\(110)_2 &= 6 \\(1100)_2 &= 12 \\(11000)_2 &= 24\end{aligned}$$

$\swarrow * 8$ 3 times left shift

#Q. Consider the given C-code and its corresponding assembly code, with a few operands U1-U4 being unknown. Some useful information as well as the semantics of each unique assembly instruction is annotated as inline comments in the code. The memory is byte-addressable.

//C-code

```
int a[10], b[10], i;
// int is 32-bit → 4B
for (i=0; i<10; i++)
a[i] = b[i] * 8;
```

$a[0] = b[0] * 8$
 $a[1] = b[1] * 8$
 $\vdots$

;assembly-code (; indicates comments)

;r1-r5 are 32-bit integer registers

;initialize r1=0, r2=10

;initialize r3, r4 with base address of a, b

L01: jeq r1, r2, end

L02: lw r5, 0(r4)

L03: shl r5, r5, U1

L04: sw r5, 0(r3)

L05: add r3, r3, U2

L06: add r4, r4, U3

L07: add r1, r1, 1

L08: jmp U4

L09: end

;if(r1==r2) goto end

;r5 ← Memory[r4+0]

;r5 ← r5 << U1

;Memory[r3+0] ← r5

;r3 ← r3 + U2

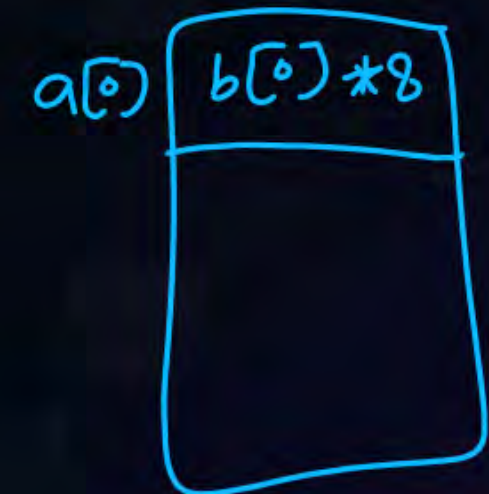
;goto U4

$$r_1 = \phi \quad 1$$

$$r_3 = a + 4$$

$$r_4 = b + 4$$

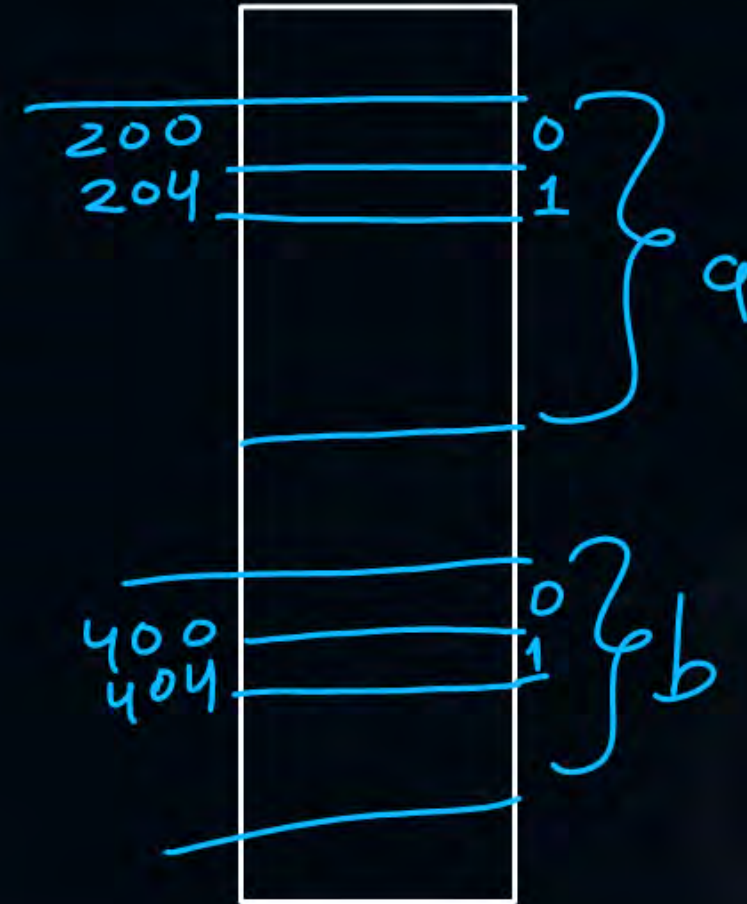
$$r_5 = b[0] * 8$$



Which of the following options is a correct replacement for operands in the position (U1, U2, U3, U4) in the above assembly code?

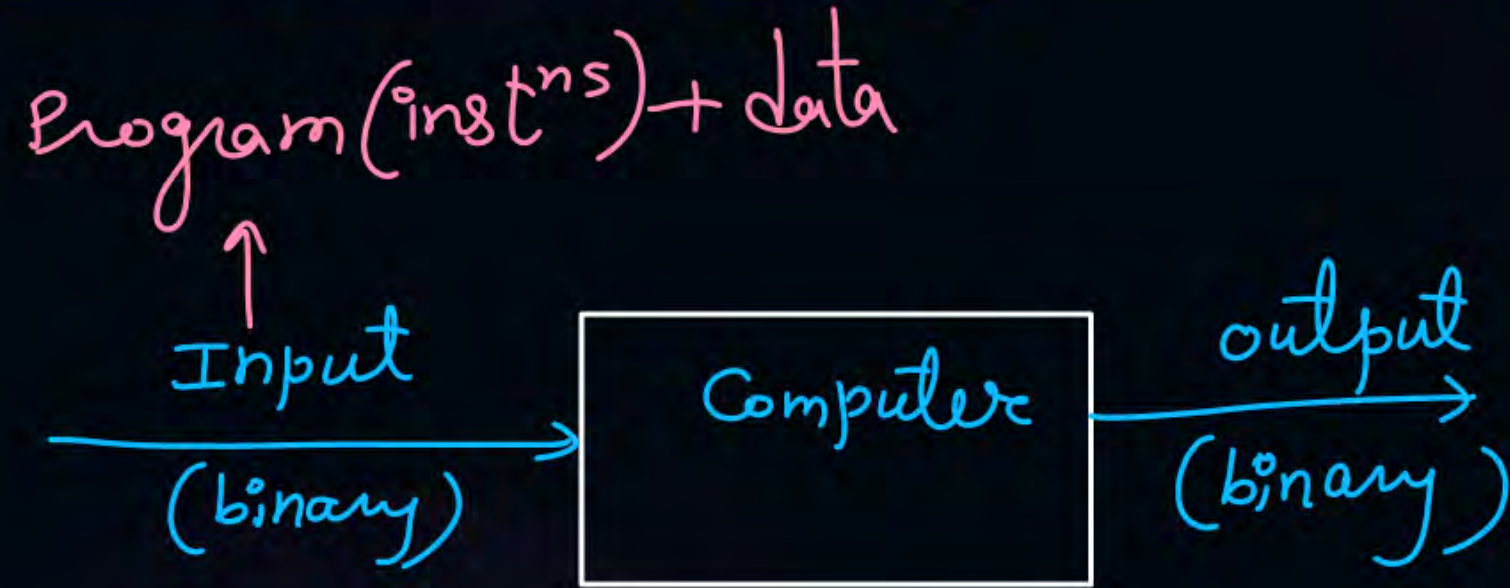
- A** (8, 4, 1, L02)
- B** ✓ (3, 4, 4, L01)
- C** (8, 1, 1, L02)
- D** (3, 1, 1, L01)

$r3 = 200$
 $r4 = 400$





Topic : Digital Computer





Topic : Instruction

```
#include<stdio.h>

void main()
{
int a, b, c;
printf("Enter 2 values: ");
scanf("%d %d", &a, &b);
c = a + b;
printf("Sum = %d", c);
}
```




Topic : Instruction

High level lang.

```
#include<stdio.h>
```

```
void main()
```

```
{
```

```
int a, b, c;
```

```
printf("Enter 2 values: ");
```

```
scanf("%d %d", &a, &b);
```

```
c = a + b;
```

```
printf("Sum = %d", c);
```

```
}
```

Programming
stmt's

(Compiler)

Language Translation

Instructions

Low level lang. prog.

(machine code or

binary code

or

byte code

or

object code)

1 0 1 1 1 0 0 0

1 0 0 0 0 0 0 1

1 1 1 1 0 0 1 0

0 1 0 1 0 1 0 1

1 1 1 1 0 1 1 0

0 1 0 1 0 1 0 1

1 0 0 0 1 1 1 1

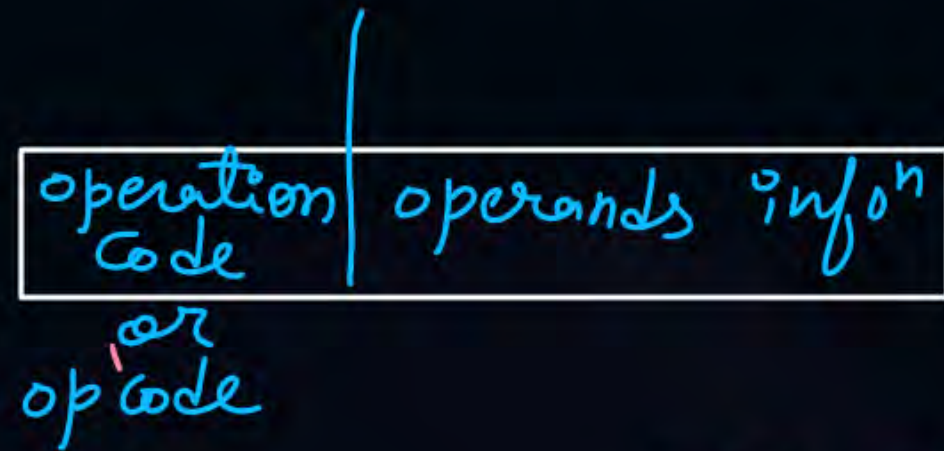
1 0 1 0 0 0 1 1

0 0 1 1 1 1 0 1



Topic : Instruction

A group of bits which instructs computer to perform some operation



000 $\Rightarrow$ ADD
001 $\Rightarrow$ SUB
010 $\Rightarrow$ complement
.
.
.
111 $\Rightarrow$ —

if no. of distinct operations CPU can perform = 8

$\Downarrow$

instⁿ op code = 3 bits

$\rightarrow$ no. of distinct inst^{ns} supported by CPU.



Topic : Instruction



no. of ^{distinct} inst^{ns} supported by CPU = n

$$\text{opcode} = \lceil \log_2 n \rceil \text{ bits}$$



Topic : ISA

→ instⁿ set architecture

→ collection of all inst^{ns} supported by a CPU.

size of ISA
or
size of instⁿ set } ⇒ no. of distinct inst^{ns} supported by a CPU.

#Q.

Q.28	Which of the following is/are part of an Instruction Set Architecture of a processor?
(A) ✗	The size of the cache memory
(B) ✗	The clock frequency of the processor
(C) ✗	The number of cache memory levels
(D) ✓	The total number of registers



Topic : Types of Instruction

- 3-Address Instruction:
- 2-Address Instruction:
- 1-Address Instruction:
- 0-Address Instruction:



Topic : 3-Address Instruction

Max 3 addresses can be specified within an instruction for operands

OpCode	Add. 1	Add. 2	Add. 3
--------	-----------	-----------	-----------

ex:-

10101	11	10	01
↓	↓	↓	↓
ADD	R3, R2, R1		

$R3 \leftarrow R2 + R1$ or
default

$R3 + R2 \rightarrow R1$



Topic : 2-Address Instruction

Max 2 addresses can be specified within an instruction

opcode	add. 1	add. 2
--------	-----------	-----------

10101	11	10
↓	↓	↓
ADD	R3, R2	

$R3 \leftarrow R3 + R2$ or $R3 + R2 \rightarrow R2$
default



Topic : 1-Address Instruction

Max 1 address can be specified within an instruction

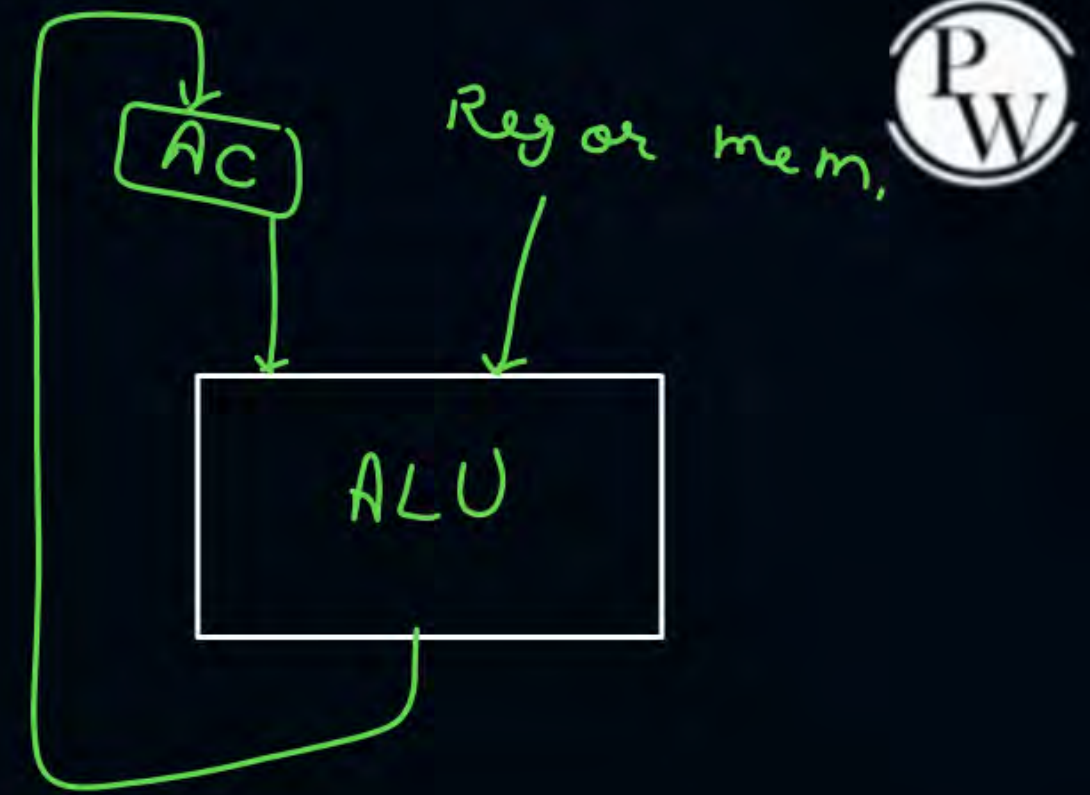
opcode	add. 1
--------	--------

10101 11

↓ ↓
ADD R3

⇓
AC-based arch.

$$AC \leftarrow AC + R3$$





Topic : 0-Address Instruction

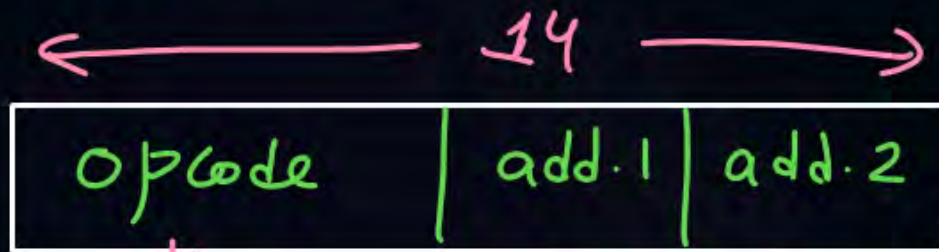


No any address can be specified within an instruction

opcode

#Q. Consider a digital computer which supports only 2-address instructions each with 14-bits. If address length is 5-bits then maximum and minimum how many instructions the system can support?

2-add. Inst^{ns}



$$= 14 - (5 + 5)$$

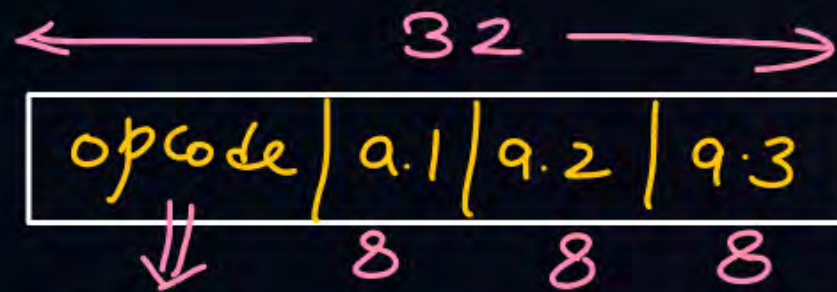
$$= 4 \text{ bits}$$

$$\rightarrow \text{max no. of inst}^{\text{ns}} \text{ supported} = 2^4 = 16$$

$$\text{min} \quad || \quad = 1$$

#Q. Consider a digital computer which supports only 3-address instructions each with 32-bits. If address length is 8-bits then maximum and minimum how many instructions the system can support?

3-add. inst's



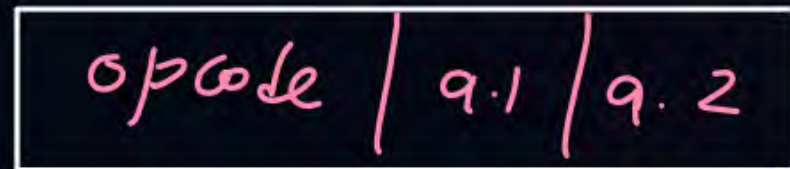
8 bits

$$\begin{aligned} \max &= 2^8 = 256 \\ \min &= 1 \end{aligned}$$

#Q. Consider a digital computer which supports 64 2-address instructions. If address length is 9-bits then the length of the instruction is 24 bits?

no. of inst^{ns} supported = 64 $\Rightarrow$ opcode = 6 bits

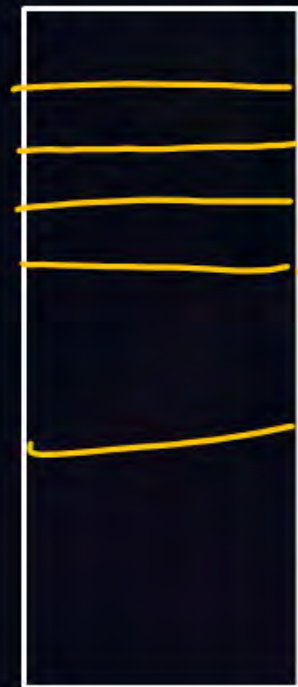
2-add. inst^{ns}



6 9 9
└──────────┘
24 bits = 3 bytes

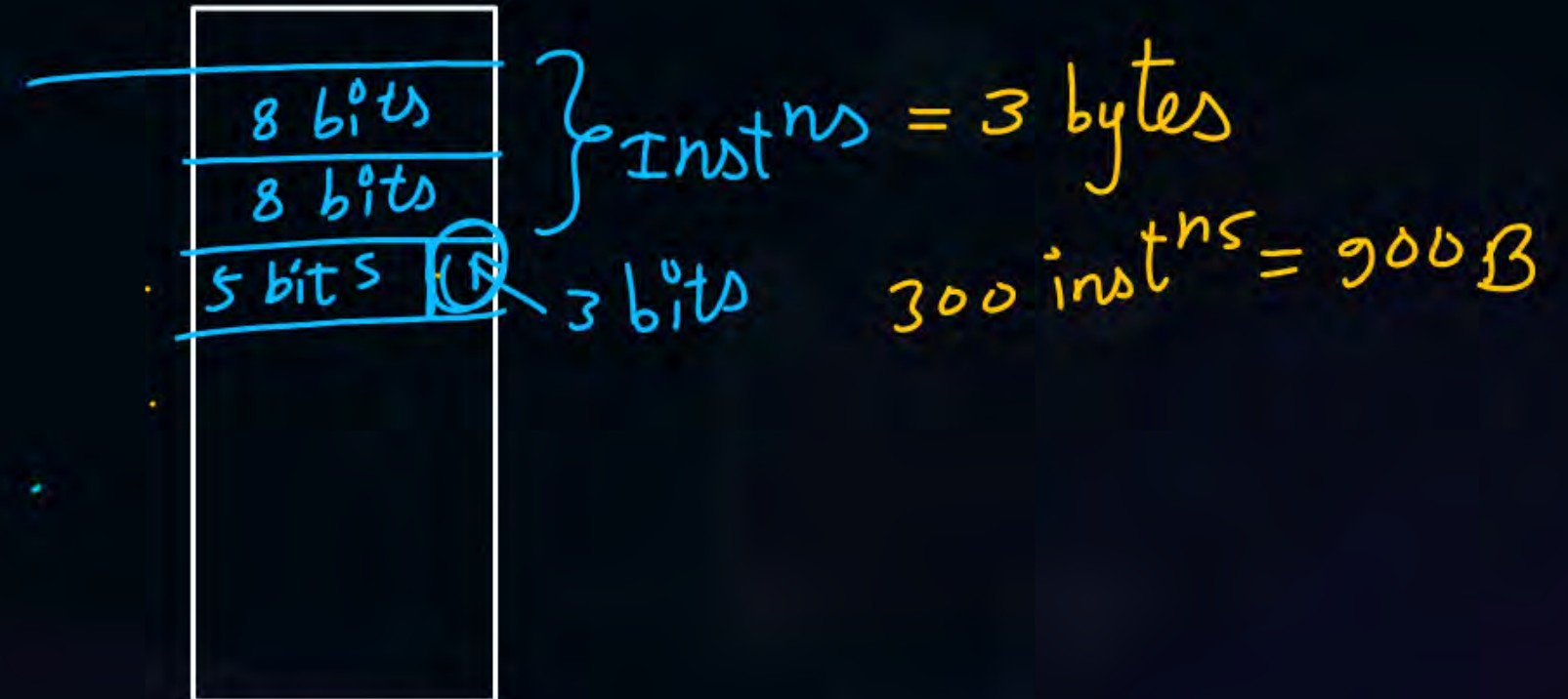
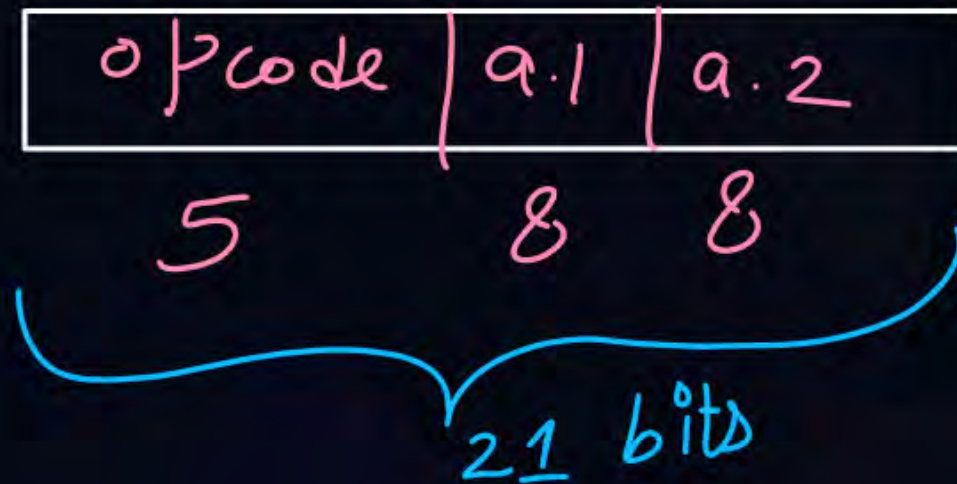
#Q. Consider a digital computer which supports 64 2-address instructions. If address length is 9-bits then the length of the instruction is 24 bits?

In above question: Each instruction must be stored in memory in a byte-aligned fashion. If a program has 200 instructions, then amount of memory required to store the program text is ____ bytes?

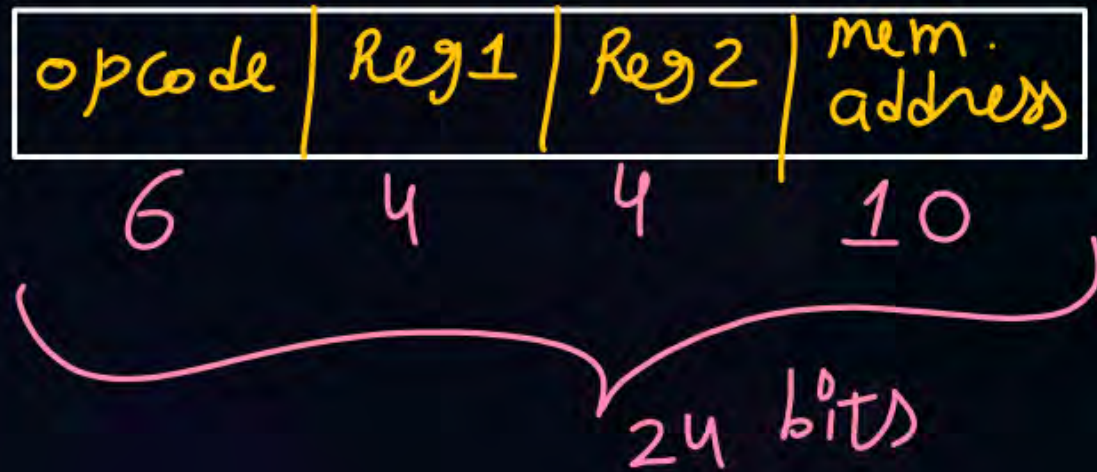


Prog.
 $200 \text{ inst}^{\text{ns}} = 200 * 3 = 600 \text{ B}$

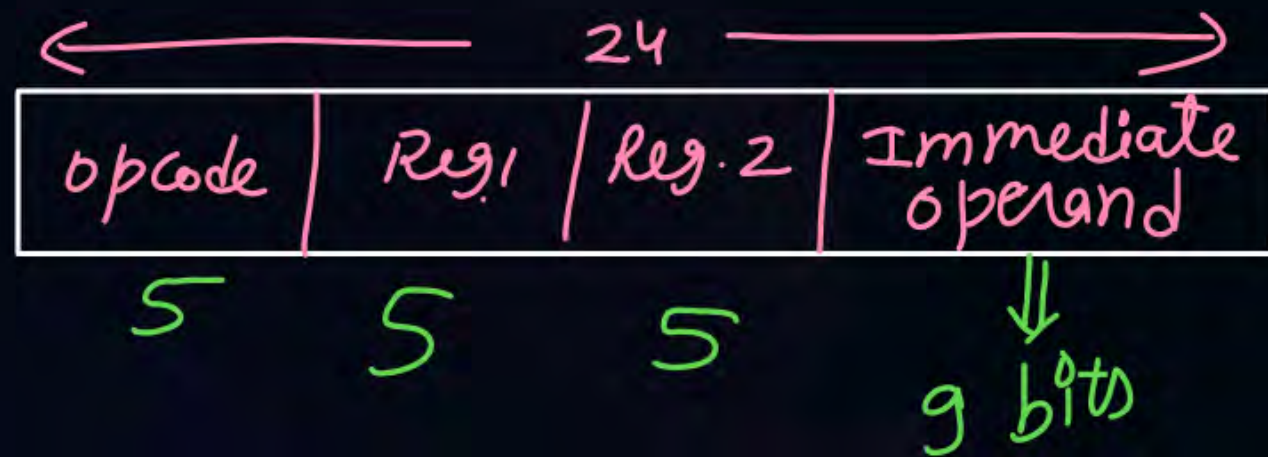
- #Q. Consider a digital computer which supports 32 2-address instructions. Consider the address length is 8-bits. Each instruction must be stored in memory in a byte-aligned fashion. If a program has 300 instructions, then amount of memory required to store the program text is 900 bytes?



- #Q. A processor has 50 distinct instructions and 16 general purpose registers. Each instruction in system has one opcode field, 2 register operand field and a 10 bits memory address field. The length of the instruction is 24 bits?
- Handwritten notes:*
→ opcode = 6 bits
→ Reg. number = 4 bits



- #Q. A processor has 20 distinct instructions and 32 general purpose registers. A 24 bits instruction word has an opcode, two register operands and an immediate operand. The number of bits available for the immediate operand field is 9?



#Q. A processor has 20 distinct instructions and 32 general purpose registers. A 24-bit instruction word has an opcode, two register operands and an immediate operand. The number of bits available for the immediate operand field is 9?

In above question: Assume that immediate operand field is an unsigned number, What is its maximum and minimum value possible?

$$\Rightarrow \begin{aligned} \max &= 2^9 - 1 = 511 \\ \min &= 0 \end{aligned}$$

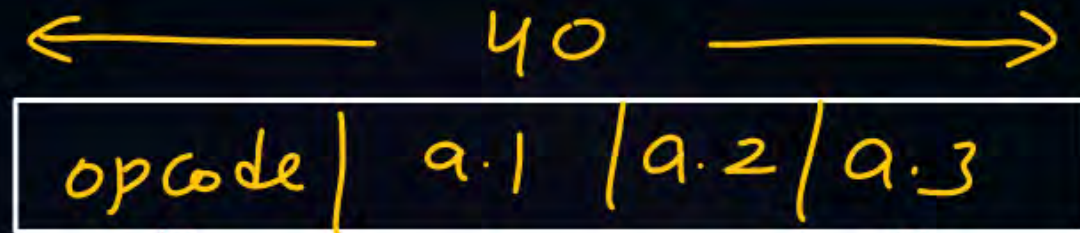
	min	max
Unsigned	0	$(2^n - 1)$
sign mag.	$-(2^{n-1} - 1)$	$+(2^{n-1} - 1)$
1 st comp.	-2^{n-1}	$+(2^{n-1} - 1)$
2 nd comp.		

[NAT]



byte addressable

#Q. Consider a system which support only 3 address instructions only, and supports 256B memory. If the instruction size is 40-bits then maximum & minimum number of instruction supported by the system are?



16 bits

$$\begin{aligned} \text{max} &= 2^{16} \\ \text{min} &= 1 \end{aligned}$$

$$\text{no. of cells} = \frac{256 \text{ B}}{1 \text{ B}} = 256 = 2^8$$

add. = 8 bits

True/False

$R1 \leftarrow R2 + M[2000]$

$R3 \leftarrow R4 + R5$

$\rightarrow$ addition $\Rightarrow$ Same opcode

opcode defines instⁿ

Above two instructions treated as same type when counted into Instruction set architecture? *True*

True/False

False

32-bit architecture CPU has 1 word instruction, it means instruction size = 5B?

↓
32 bits = 4B

True/False

False

X-bit architecture computer means the address bus width is x-bits?

no relation



2 mins Summary



Topic

Instructions

Topic

Opcode



Happy Learning

THANK - YOU